

mccode-dev / McCode

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McStas sample functionality matrix

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Peter Willendrup edited this page 2 minutes ago · [32 revisions](#)

McStas sample model functionality-matrix



( Original master version was available at the URL <https://confluence.esss.lu.se/display/MCSTAS/McStas+sample+model+functionality-matrix> )

Status of McStas sample components relevant for McStas 3.5.x

|  | McStas sample comp<br>+ author info in italic | Model<br>description | Main use<br>areas |
|--|-----------------------------------------------|----------------------|-------------------|
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|   |                                                                                                   |                                                                                                                        |                                          |
|---|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1 | <a href="#">Incoherent</a> (Vanadium, Plexiglass etc.)<br><br><a href="#">McStas team</a>         | Simple incoherent scatterer                                                                                            | Generic, imaging                         |
| 2 | <a href="#">Tunelling_sample</a><br><br><a href="#">McStas team</a> / <a href="#">Kim Lefmann</a> | Idem 1, plus tunneling peaks and QE broadening                                                                         | Quasi-elastic scattering, backscattering |
| 3 | <a href="#">PowderN</a><br><br><a href="#">McStas team</a> / <a href="#">Peter Willendrup</a>     | Debye-scherrer cones, tabular input ( <a href="#">lau</a> / laz)                                                       | Powder diffraction, imaging              |
| 4 | <a href="#">Sample_nxs</a><br><br><a href="#">Mirko Boin, HZB</a>                                 | Debye-scherrer cones, unit-cell / atom input list                                                                      | Powder diffraction, (future: imaging)    |
| 5 | <a href="#">Single_crystal</a><br><br><a href="#">McStas team</a>                                 | Bragg spots, tabular input ( <a href="#">lau</a> ). "Perfect imperfect"<br><br>single crystal with mosaicity / lattice | Single crystal and MX diffraction        |

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<https://github.com/mccode-dev/McCode/wiki/McStas-sample-functionality-matrix>

|    |                                                                                                                                                                           |                                                                                                  |                                        |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------|
|    |                                                                                                                                                                           | variation                                                                                        |                                        |
| 6  | <a href="#">Sans_spheres</a><br>( <a href="#">and other similar</a> )<br><a href="#">McStas team</a> and<br><a href="#">Martin Cramer</a><br><a href="#">Pedersen, KU</a> | Hard spheres in thin solution<br>and other<br><br>models, defined per-<br>component...           | SANS                                   |
| 7  | <a href="#">SANS_benchmark2</a><br><a href="#">Heinrich Frielinghaus,</a><br><a href="#">FZJ/JCNS</a>                                                                     | Experimentally-<br>benchmarked model<br><br>set for SANS                                         | SANS                                   |
| 8  | <a href="#">SASview_models</a><br><a href="#">McStas team</a>                                                                                                             | "Any" model from SASview /<br>SASmodels                                                          | SANS                                   |
| 9  | <a href="#">Multilayer_sample</a><br><a href="#">Rob Dalglish, ISIS</a><br><a href="#">STFC</a>                                                                           | Multilayer-sample (additions<br>of phase via matrix-<br>formalism) with incoherent<br>background | Reflectometry                          |
| 10 | <a href="#">Phonon_simple</a><br><a href="#">McStas team</a> / <a href="#">Kim</a><br><a href="#">Lefmann</a>                                                             | Single-branch acoustic<br><br>phonon in FCC lattice                                              | Inelastic<br>scattering<br><br>phonons |
|    |                                                                                                                                                                           |                                                                                                  |                                        |

|    |                                                                                               |                                                                                                                                                |                                                                                    |
|----|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 11 | <a href="#">Isotropic_Sqw</a><br><a href="#">McStas team</a> / <a href="#">Emmanuel Farhi</a> | Structure and dynamics<br>in isotropic materials<br>(liquids, powders etc.)                                                                    | Inelastic<br>scattering><br>diffraction,<br><br>isotropic<br>materials,<br>imaging |
| 12 | <a href="#">Res_sample</a><br><a href="#">McStas team</a>                                     | Resolution-oriented<br>sample component                                                                                                        | Generic                                                                            |
| 13 | <a href="#">TOFRes_sample</a><br><a href="#">McStas team</a> / <a href="#">Kim Lefmann</a>    | Idem Res_sample, with TOF<br>support                                                                                                           | Generic                                                                            |
| 14 | <a href="#">Spot_sample</a><br><a href="#">Garrett Granroth</a> ,<br>SNS/ORNL                 | Resolution-oriented<br>sample component<br><br>Dirac delta-functions in (Q<br>and energy)                                                      | Inelastic<br>scattering>                                                           |
| 15 | <a href="#">Union components</a> ,<br><a href="#">Mads Bertelsen, ESS</a>                     | A set of components that<br>allows to build a complex<br>sample/sample environment<br>from basic geometries and<br>physics/material properties | Generic                                                                            |

|    |                                                                                                                |                                                                                                                       |                                                                     |
|----|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 16 | <a href="#">Single_crystal_inelastic</a><br><a href="#">Duc Le, ISIS STFC</a>                                  | 4D-equivalent of<br>Isotropic_Sqw /<br>Single_crystal                                                                 | Elastic and<br>inelastic<br><br>experiments<br>with<br><br>crystals |
| 17 | <a href="#">Magnon_bcc</a><br><a href="#">McStas team</a> / <a href="#">Kim Lefmann</a>                        | FM / AFM magnon in BCC<br>lattice                                                                                     | Inelastic<br>scattering<br>magnon                                   |
| 18 | <a href="#">NCrystal_sample</a><br><a href="#">Xiao Xiao Cai, DTU</a><br><a href="#">Nutech/ESS</a>            | Single crystal and powder<br>diffraction, with isotropic<br>inelastic scattering                                      | Powder and<br>Single_crystal<br>diffraction,<br>imaging             |
| 19 | <a href="#">Single_magnetic_crystal</a><br><a href="#">Linda Udby</a> KU, + <a href="#">Erik Knudsen</a> , DTU | Bragg spots from lattice ala<br>Single_crystal<br><br>plus magnetic lattice.<br>Tabular input ( <a href="#">lau</a> ) | Single crystal<br>magnetic<br>diffraction                           |
| 20 | "Specular<br>reflectometry" (via<br><a href="#">Mirror.comp</a> )                                              | Use a reflectivity-curve with<br>e.g. <a href="#">Mirror.comp</a>                                                     | Reflectometry                                                       |
|    |                                                                                                                |                                                                                                                       |                                                                     |

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|----|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|
| 21 | Below this line not yet available in repo                                                     | Below this line not yet available in repo                       | Below this line not yet available in repo |
| 22 | "Polycrystal"<br><a href="#">Alberto Cereser</a> + <a href="#">Erik Knudsen</a> , DTU Physics | Engineering-diffraction / imaging oriented<br>multigrain sample | Engineering-diffraction / imaging         |

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